



WHAT'S NEW IN DOLFINY - HIGH-LEVEL CONVENIENCE WRAPPERS FOR DOLFINx

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Dolfiny is a collection of convenience wrappers and extended functionality for DOLFINx, first presented in FEniCS 2021 [1] and continuously developed. Over time, the number of features has grown and covers wrappers for PETSc SNES and SLEPc, static condensation, arc-length load control, ODE integration, function space restriction and other minor helpers. More recently we have added support for physical units [2], dimensional analysis [3] and a PETSc TAO wrapper [4]. In this poster contribution we provide an overview of its capabilities and plans for the future.

References

1. Zilian, A., Habera, M. (23 March 2021). dolfiny: Convenience wrappers for DOLFINx. FEniCS 2021, Cambridge, United Kingdom.
2. Habera, M., Zilian, A. (18 June 2025). Physical units in UFL. FEniCS 2025, Groningen, Netherlands.
3. Zilian, A., Habera, M. (20 June 2025). Dimensional analysis in UFL. FEniCS 2025, Groningen, Netherlands.
4. Kühner, P. T., Habera, M., Zilian, A. (18 June 2025). PETSc TAO support for optimisation problems with FEniCSx. FEniCS 2025, Groningen, Netherlands.